$\boxed{\text{Soln:}}$ $S(x)$: x is a student of this class $C(x)$: x has studied calculus.

Domain: All students of the school

$$\boxed{\forall x(S(x) \to C(x))}$$

Finally, when we are interested in the background of people in subjects besides calculus, we may prefer to use the two-variable quantifier $Q(x,y)$ for the statement "Student x has studied subject y." Then we would replace $C(x)$ by $Q(x, \text{calculus})$ to obtain $\boxed{\forall x(S(x) \to Q(x, \text{calculus}))}$

$\boxed{\text{Example 24:}}$ Express the statements "Some student in this class has visited Mexico" and "Every student in this class has visited either Canada or Mexico" using predicates and quantifiers.

$\boxed{\text{Soln:}}$ $S(x)$: x is a Student of this ~~mexic~~ class Domain: Students of a school

$M(x)$: x has visited Mexico

$C(x)$: x has visited Canada

Some student in this class has visited Mexico $\to \exists x(S(x) \land M(x))$

Every student in this class has visited either Canda or Mexico $\to$

$$\boxed{\forall x(S(x) \to M(x) \lor C(x))}$$

<u>Examples from Lewis Carroll</u>

Lewis Carroll (really C.L. Dodgson writing under a pseudonym), the author of $\boxed{\text{Alice in Wonderland}}$, is also the author of several works on symbolic logic. His books contain many examples of reasoning using quantifiers. Examples 26 and 27 come from his book $\boxed{\text{Symbolic Logic}}$; other examples from that book are given in the exercises at the end of this section.

$\boxed{\text{Example 26:}}$ Consider these statements. The first two are called $\boxed{\text{premises}}$ and the third is called the $\boxed{\text{conclusion}}$. The entire set is called an $\boxed{\text{argument}}$.

- "All lions are fierce."
- "Some lions do not drink coffee".
- "Some fierce creatures do not drink coffee."